

Library Management System

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Console-Based Database Management Application

Purpose:

This project was developed to demonstrate the implementation of a simple library management system using Python and MySQL. The application supports administrator operations such as user management, book management, and search functionality.

A. User Login

This screen demonstrates secure user authentication before accessing the library application features.

Login options are displayed

```
=== Library Management System ===
1. Admin Login
2. User Login
3. Exit
Enter choice: |
```

User chooses 'User Login' and enters Userid and Password.

```
=== Library Management System ===
1. Admin Login
2. User Login
3. Exit
Enter choice: 2
Enter user credentials:
User ID: 1
Password:
User login successful!
```

Functionalities Demonstrated

- Username verification
- Password authentication
- Protected password entry using 'getpass' module of Python
- Role-based system access

B. Display User Menu

Post login the menu options for the User are displayed

```
User login successful!

--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: |
```

Functionalities Demonstrated

- Database retrieval
- Structured Result display

C. Display All Books

The user can view all books and associated details stored in the database using the 'Display All Books' feature.

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 2

List of Books:
ID: 1 Title: The Great Gatsby Author: F. Scott Fitzgerald Year: 1984 Copies: 3
ID: 2 Title: Nineteen Eighty-four Author: George Orwell Year: 1949 Copies: 2
ID: 3 Title: The Old Man and the Sea Author: Ernst Hemingway Year: 1952 Copies: 3
ID: 4 Title: Sonar Kella Author: Satyajit Ray Year: 1971 Copies: 4
ID: 5 Title: Gitanjali Author: Rabindranath Tagore Year: 1926 Copies: 5
ID: 6 Title: The Stranger Author: Albert Camus Year: 1970 Copies: 6
ID: 7 Title: For Whom the Bell Tolls Author: Ernst Hemingway Year: 1956 Copies: 7
```

Functionalities Demonstrated

- Database retrieval
- Structured Result display

D. Search Book(s)

This feature allows the user to locate book(s) using the book id or keyword/ title search. Partial keyword/ title search is allowed.

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 1
Enter book title or ID to search: Old
ID: 3 Title: The Old Man and the Sea Author: Ernst Hemingway Year: 1952 Copies: 3
```

Functionalities Demonstrated

- Search query handling

Library Management System

- Partial Search
- Structured Result display

E. Issue Book

First look up all books available in the system using the 'Display All Books' feature. (Note the count corresponding to Book ID : 1 is 3 before the user issues it.)

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 2

List of Books:
ID: 1 Title: The Great Gatsby Author: F. Scott Fitzgerald Year: 1984 Copies: 3
ID: 2 Title: Nineteen Eighty-four Author: George Orwell Year: 1949 Copies: 2
ID: 3 Title: The Old Man and the Sea Author: Ernst Hemingway Year: 1952 Copies: 3
ID: 4 Title: Sonar Kella Author: Satyajit Ray Year: 1971 Copies: 4
ID: 5 Title: Gitanjali Author: Rabindranath Tagore Year: 1926 Copies: 5
ID: 6 Title: The Stranger Author: Albert Camus Year: 1970 Copies: 6
ID: 7 Title: For Whom the Bell Tolls Author: Ernst Hemingway Year: 1956 Copies: 7
```

User issues a Book ID : 1 in her/ his name using the 'Issue Book' feature

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 3
Enter Book ID to issue: 1
Book ID 1 issued to User ID 1 on 2026-05-28.
```

After issuance, note that the count of books corresponding to Book ID : 1 is reduced by 1

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 2

List of Books:
ID: 1 Title: The Great Gatsby Author: F. Scott Fitzgerald Year: 1984 Copies: 2
ID: 2 Title: Nineteen Eighty-four Author: George Orwell Year: 1949 Copies: 2
ID: 3 Title: The Old Man and the Sea Author: Ernst Hemingway Year: 1952 Copies: 3
ID: 4 Title: Sonar Kella Author: Satyajit Ray Year: 1971 Copies: 4
ID: 5 Title: Gitanjali Author: Rabindranath Tagore Year: 1926 Copies: 5
ID: 6 Title: The Stranger Author: Albert Camus Year: 1970 Copies: 6
ID: 7 Title: For Whom the Bell Tolls Author: Ernst Hemingway Year: 1956 Copies: 7
```

Functionalities Demonstrated

- User input handling
- Database insertion & update

F. My Issued Books

User can view the books that have been issued to her/him using the 'My Issued Books' feature. The function shows the user the history all books issued and returned.

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 5
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-27 Returned on: 2026-05-27
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-28 Returned on: 2026-05-28
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-28 Returned on: 2026-05-28
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-28 Returned on: 2026-05-28
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-28 Returned on: 2026-05-28
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-28 Returned on: 2026-05-28
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-28 Returned on: None
```

Functionalities Demonstrated

- Database retrieval
- Structured Result display

G. Return Book

Allows the user to return the book that had been issued to her/him using the 'Return Book' feature.

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 4
Enter Book ID to return: 1
Book ID 1 returned by User ID 1 on 2026-05-28.
```

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Now, display that the book was actually returned using the 'My Issued Books' feature.

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 5
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-27 Returned on: 2026-05-27
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-28 Returned on: 2026-05-28
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-28 Returned on: 2026-05-28
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-28 Returned on: 2026-05-28
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-28 Returned on: 2026-05-28
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-28 Returned on: 2026-05-28
Book ID: 1 Title: The Great Gatsby Issued on: 2026-05-28 Returned on: 2026-05-28
```

After return, note that the count of books corresponding to Book ID : 1 is increased by 1

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 2

List of Books:
ID: 1 Title: The Great Gatsby Author: F. Scott Fitzgerald Year: 1984 Copies: 3
ID: 2 Title: Nineteen Eighty-four Author: George Orwell Year: 1949 Copies: 2
ID: 3 Title: The Old Man and the Sea Author: Ernst Hemingway Year: 1952 Copies: 3
ID: 4 Title: Sonar Kella Author: Satyajit Ray Year: 1971 Copies: 4
ID: 5 Title: Gitanjali Author: Rabindranath Tagore Year: 1926 Copies: 5
ID: 6 Title: The Stranger Author: Albert Camus Year: 1970 Copies: 6
ID: 7 Title: For Whom the Bell Tolls Author: Ernst Hemingway Year: 1956 Copies: 7
```

Functionalities Demonstrated

- User input handling
- Database insertion & update

H. Add Note

User can add notes using the 'Add Note' feature.

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 6
Enter note content: I loved the book The Great Gatsby
Note added successfully with ID 5
```

Functionalities Demonstrated

- User input handling
- Database insertion

I. Display My Notes

User can display the notes entered by her/him using the 'Display All Notes' feature.

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 9
Notes for User ID 1:
ID: 1 Date: 2026-05-27 Content: Read chapters 1-3
ID: 3 Date: 2026-05-28 Content: Testing Note feature
ID: 5 Date: 2026-05-28 Content: I loved the book The Great Gatsby
```

Functionalities Demonstrated

- Database retrieval
- Structured Result display

J. Update Note

User can update a note entered by her/him using the 'Update Note' feature.

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 8
Enter Note ID to update: 1
Current content: Read chapters 1-3
Enter new content: Read chapter 1-3 in the first week and then chapters 4-8 in the next week. Loving it!
Note updated successfully.
```

User can look up the updated note using the 'Display My Notes' feature

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 9
Notes for User ID 1:
ID: 1 Date: 2026-05-27 Content: Read chapter 1-3 in the first week and then chapters 4-8 in the next week.
Loving it!
ID: 3 Date: 2026-05-28 Content: Testing Note feature
ID: 5 Date: 2026-05-28 Content: I loved the book The Great Gatsby
```

Functionalities Demonstrated

- User input handling
- Database update

K. Delete Note

User can delete a note entered by her/him using the 'Delete Note' feature.

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 7
Enter Note ID to delete: 3
Are you sure you want to delete note 3? (y/n): y
Note deleted successfully.
```

User can look up if the note was deleted using the 'Display My Notes' feature.

Library Management System

```
--- User Menu ---
1. Search Book
2. Display All Books
3. Issue Book
4. Return Book
5. My Issued Books
6. Add Note
7. Delete Note
8. Update Note
9. Display My Notes
10. Search Notes
11. Logout
Enter your choice: 9
Notes for User ID 1:
ID: 1 Date: 2026-05-27 Content: Read chapter 1-3 in the first week and then chapters 4-8 in the next week.
Loving it!
ID: 5 Date: 2026-05-28 Content: I loved the book The Great Gatsby
```

Functionalities Demonstrated

- User input handling
- Database deletion